

Tutorial APACHE

1. Preparar y actualizar Ubuntu

Actualizarla información disponible con (*sudo apt update*)

```
david@Despliegue:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://es.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://es.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://es.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:4 http://es.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [714 kB]
Get:5 http://es.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [167 kB]
Get:6 http://es.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [98.4 kB]
Get:7 http://es.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [98.4 kB]
Get:8 http://es.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [167 kB]
Get:9 http://es.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [3
```

2. Instalar Apache2

Instalar el Apache2 con (*sudo apt install apache2 -y*)

```
david@Despliegue:~$ sudo apt install apache2 -y
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30      linux-image-unsigned-7.0.0-30-generic  linux-modules-7.0.0-30-generic
  linux-headers-7.0.0-30-generic  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  apache2

Installing dependencies:
  apache2-bin  apache2-data  apache2-utils  libapr1t64  libaprutil1-dbd-sqlite3  libaprutil1t64

Suggested packages:
  apache2-doc  apache2-suexec-pristine  | apache2-suexec-custom  www-browser

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 125
  Download size: 2,119 kB
  Space needed: 8,218 kB / 21.1 GB available
```

3. Comprobar que Apache funciona

Con (*sudo systemctl status apache2*) comprobar el estado del servicio

```
david@Despliegue:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-10-01 16:56:32 UTC; 1min 49s ago
     Invocation: 72734ad682644d3d8e563c21b244fe16
       Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 2028 (apache2)
      Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0"
        Tasks: 55 (limit: 4894)
       Memory: 5.8M (peak: 6.1M)
          CPU: 74ms
       CGroup: /system.slice/apache2.service
              └─2028 /usr/sbin/apache2 -k start -DFOREGROUND
                └─2044 /usr/sbin/apache2 -k start -DFOREGROUND
                  └─2045 /usr/sbin/apache2 -k start -DFOREGROUND

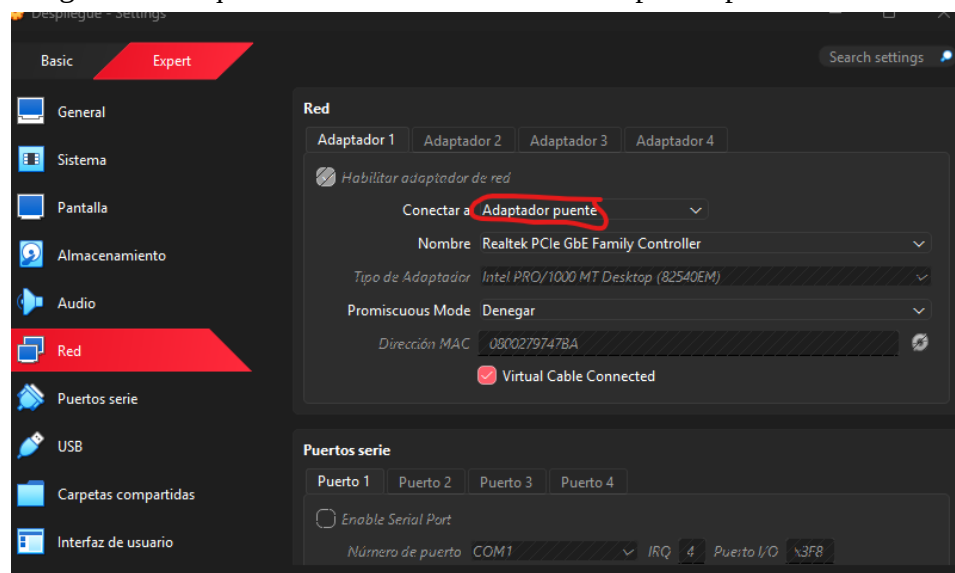
Oct 01 16:56:32 Despliegue systemd[1]: Starting apache2.service - The Apache HTTP Server.
Oct 01 16:56:32 Despliegue apache2[2028]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, please see the documentation for instructions on how to set it properly
Oct 01 16:56:32 Despliegue systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-18/18 (END)
```

4. Probar Apache desde el navegador

Con (*ip a*) compruebo la ip de la máquina virtual q es:

```
david@Despliegue:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP
   link/ether 08:00:27:97:47:ba brd ff:ff:ff:ff:ff:ff
   altname enx0800279747ba
   inet 192.168.88.99/24 metric 100 brd 192.168.88.255 scope global dynamic enp0s3
       valid_lft 585sec preferred_lft 585sec
   inet6 fe80::a00:27ff:fe97:47ba/64 scope link proto kernel ll
       valid_lft forever preferred_lft forever
david@Despliegue:~$
```

Luego en la maquina virtual cambiar la red a adaptador puente



Para terminar poner en el navegador (<http://192.168.88.99>)



This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
- They are activated by symlinking available configuration files from their respective `*-available/` counterparts. These should be managed by using our helpers `a2enmod`, `a2dismod`, `a2ensite`,

5. Crear una página web propia

Compruebo el contenido de `/var/www/html/` que es la carpeta habitual para el sitio web predeterminado

```
david@Despliegue:~$ ls -l /var/www/html
total 12
-rw-r--r-- 1 root root 10672 Oct  1 16:56 index.html
david@Despliegue:~$ _
```

Luego hacer una copia de seguridad con (***sudo cp /var/www/html/index.html /var/www/html/index.html.bask***)

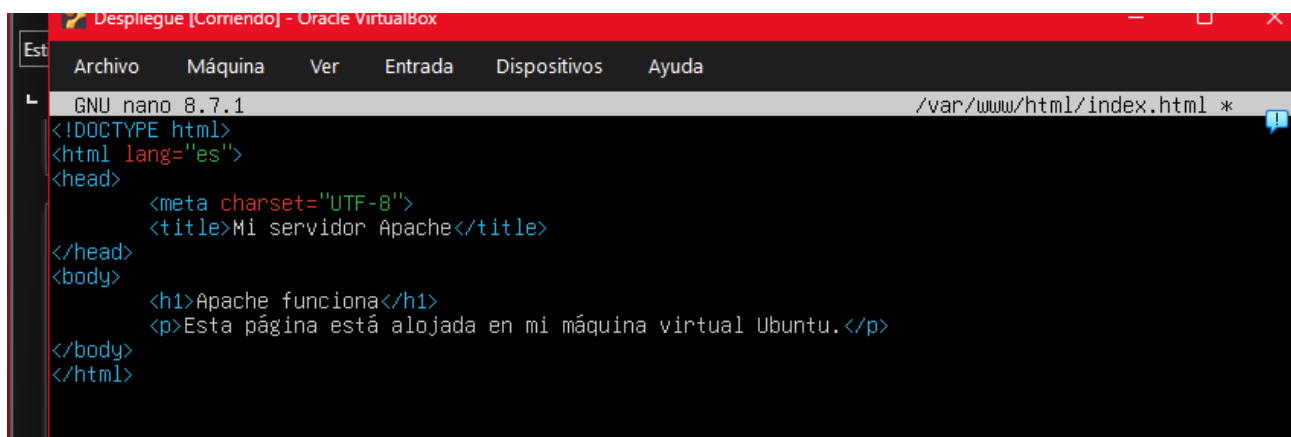
```
david@Despliegue:~$ sudo cp /var/www/html/index.html /var/www/html.bask
david@Despliegue:~$
```

y despues abrirlo (***sudo nano /var/www/html/index.html***)

```
david@Despliegue:~$ sudo nano /var/www/html/index.html_
```

Sustituyo el contenido por este ejemplo:

```
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="UTF-8">
  <title>Mi servidor Apache</title>
</head>
<body>
  <h1>¡Apache funciona!</h1>
  <p>Esta página está alojada en mi máquina virtual Ubuntu.</p>
</body>
</html>
```



```
Despliegue [Corriendo] - Oracle VirtualBox
Est  Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda
GNU nano 8.7.1 /var/www/html/index.html *
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="UTF-8">
  <title>Mi servidor Apache</title>
</head>
<body>
  <h1>Apache funciona</h1>
  <p>Esta página está alojada en mi máquina virtual Ubuntu.</p>
</body>
</html>
```

6. Configurar el firewall (UFW)

Comprobar el estado de UFW con (***sudo ufw status***)

```
david@Despliegue:~$ sudo ufw status
Status: active
david@Despliegue:~$ _
```

Si UFW está activo, permitir HTTP por el puerto 80 con (***sudo ufw allow 80/tcp***)

```
david@Despliegue:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
```

7. Consultar la IP de la máquina virtual

(*hostname -I*) para conocer la dirección IP de Ubuntu :

```
david@Despliegue:~$ hostname -I
192.168.88.99
david@Despliegue:~$
```

8. Crear un Virtual Host

Crear la carpeta del sitio con (*sudo mkdir -p /var/www/mipagina*) y modificar el dueño del archivo (*sudo chown -R \$USER:\$USER /var/www/mipagina*)

```
david@Despliegue:~$ sudo mkdir -p /var/www/mipagina
david@Despliegue:~$ sudo chown -R $USER:$USER /var/www/mipagina
david@Despliegue:~$ _
```

Ahora a crear la página entrando en (*nano /var/www/mipagina/index.html*)

```
david@Despliegue:~$ sudo nano /var/www/mipagina/index.html
```

```
GNU nano 8.7.1 /var/www/mipagina/index.html
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="UTF-8">
  <title>Mi página</title>
</head>
<body>
  <h1>Mi página web con Apache</h1>
  <p>Virtual Host funcionando correctamente</p>
</body>
</html>_
```

Ahora crear el archivo de configuración (*sudo nano /etc/apache2/sites-available/mipagina.conf*)

```
david@Despliegue:~$ sudo nano /etc/apache2/sites-available/mipagina.conf
```

```
GNU nano 8.7.1 /etc/apache2/sites-available/mipagina.conf
<VirtualHost *:80>
    ServerName mipagina.local
    DocumentRoot /var/www/mipagina

    <Directory /var/www/mipagina>
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/mipagina_error.log
    CustomLog ${APACHE_LOG_DIR}/mipagina_access.log combined
</VirtualHost>
```

9. Activar y comprobar el Virtual Host

Activar el sitio (*sudo a2ensite mipagina.conf*)

```
david@Despliegue:~$ sudo a2ensite mipagina.conf
Enabling site mipagina.
To activate the new configuration, you need to run:
  systemctl reload apache2
david@Despliegue:~$ _
```

Comprobar la sintaxis (*sudo apache2ctl configtest*)

```
david@Despliegue:/home/david$ sudo apache2ctl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName'
is message
Syntax OK
david@Despliegue:/home/david$
```

Recargar Apache (*sudo systemctl reload apache2*)

```
david@Despliegue:/home/david$ sudo systemctl reload apache2
david@Despliegue:/home/david$
```

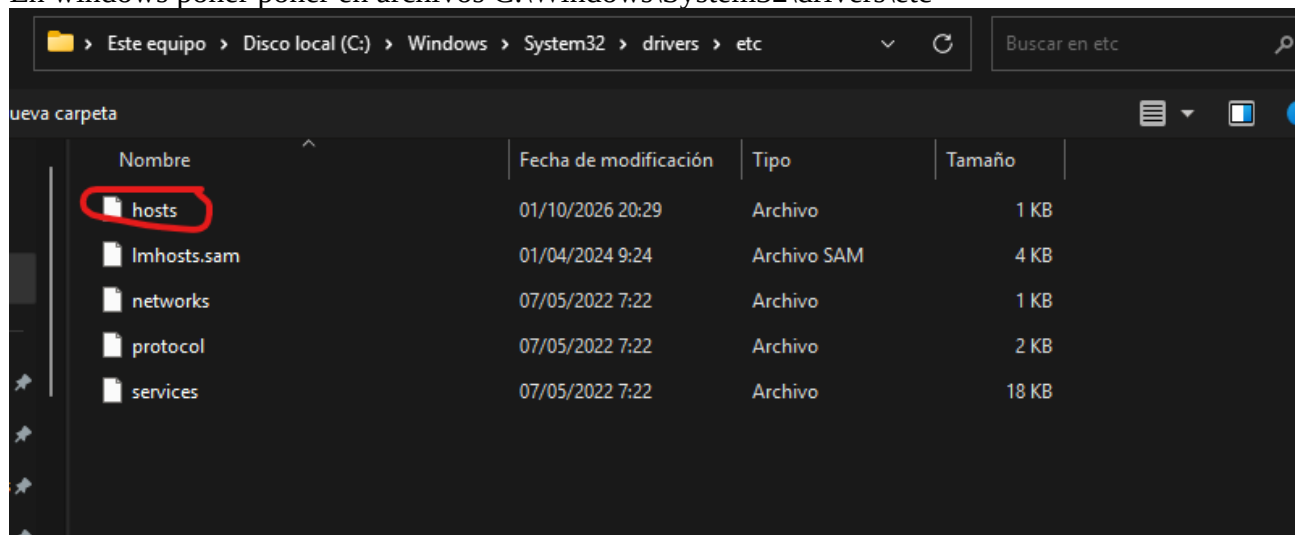
10. Asociar el nombre mediante /etc/hosts

En la maquina virtual editar el archivo hosts (*sudo nano /etc/hosts*)

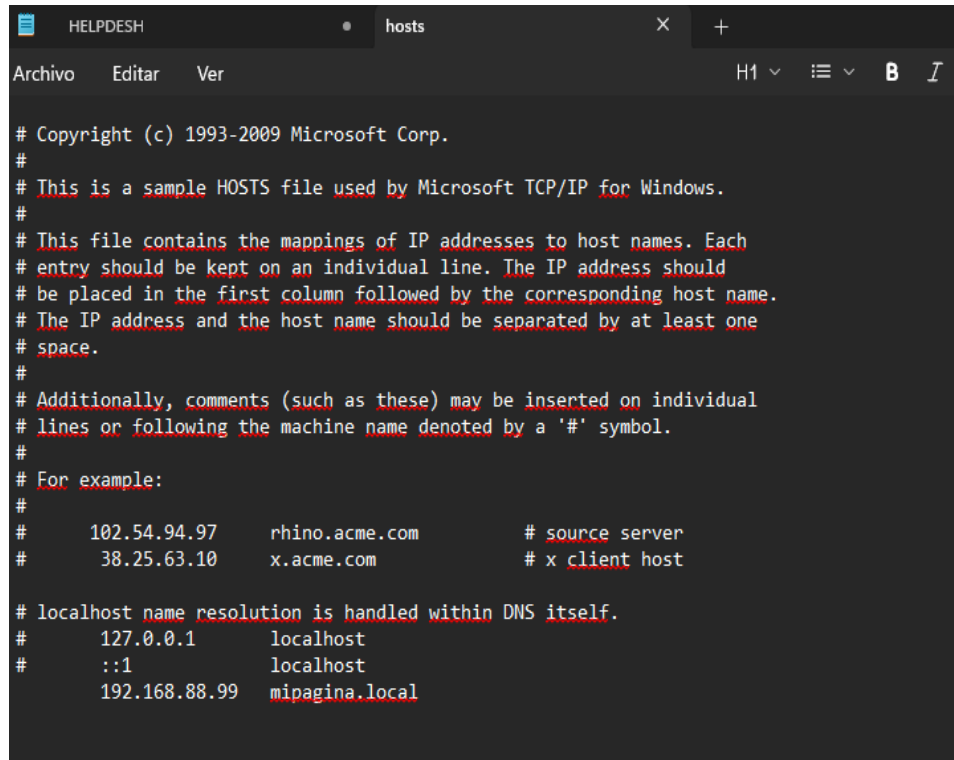
```
GNU nano 8.7.1 /etc/hosts
127.0.0.1 localhost
127.0.1.1 Despliegue
192.168.88.99 mipagina.local_

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

En windows poner poner en archivos C:\Windows\System32\drivers\etc



Y abrir host y la ultima fila poner la ip de ubuntu



```
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97       rhino.acme.com       # source server
#       38.25.63.10       x.acme.com          # x client host
#
# localhost name resolution is handled within DNS itself.
#       127.0.0.1         localhost
#       ::1               localhost
#       192.168.88.99     mipagina.local
```

Resultado:

